Maaz Qureshi

Research Engineer (ML, Embodied AI and Robotics): MASc Mechatronics University of Waterloo

Portfolio | GitHub | LinkedIn | Scholar | +1 437 870 8791 | m23qures@uwaterloo.ca | Waterloo, Canada

SKILLS

Machine Learning & Foundation Models: Vision-Language-Action models (π_0, smolVLA), World-Action Models (literature), Parameter-Efficient Fine-Tuning (PEFT: LoRA), CNNs, VLMs, Quantization (FP16/INT8), Reinforcement Learning (PPO, policy optimization), Imitation Learning (behavior cloning) & Diffusion Policy

Robotics: AR/XR (ARKit), 6D Pose Estimation, Visual-Inertial SLAM (volumetric mapping & localization), Motion Planning for 7-DoF Manipulators (IK/FK), Sensor Fusion (stereo RGB-D, LiDAR, mmWave 4D radar, IMU), 5G-Connected Factory Robotics

Languages: Python, C++, PyTorch, TensorRT (CUDA), Eigen, NumPy, ROS2

Tools: Docker, CI/CD, gRPC, NVIDIA Isaac, MuJoCo, Gazebo, RViz, Asynchronous GPU Pipeline, Debugging/Monitoring (Valgrind, Jtop)

EXPERIENCE

Research Assistant (Apple Robotics Embodied AI): May 2026 – Present

University of Waterloo, joint research with Apple Inc.

- Developed an ARKit-based egocentric pipeline on iPhone Pro, capturing 6-DoF poses, RGB-D frames, and gripper width.
- Fine-tuned Vision-Language-Action models and diffusion policies on egocentric data for embodied AI manipulation (NDA).
- Achieved model SR 100%, object move less then 20 cm from the pick point with iPhone pro and D405 camera for visual state.
- Benchmarked ARKit pose tracking against Franka Panda 6D ground truth across 5–65 °/s angular-velocity robot-arm sweeps.

Robotics Software Engineer [video] September 2025 – April 2026

Cobionix Corporation Inc. Kitchener, Canada

- Built a real-time multimodal perception pipeline at 22 FPS (SmolVLM, pose, detection) in C++/CUDA on NVIDIA Jetson.
- Fine-tuned SmolVLM via LoRA for patient-domain recognition; applied TensorRT FP16/INT8 quantization for edge inference.
- Built stereo RGB-D sensor fusion with intrinsic/extrinsic spatiotemporal calibration, achieving >90% 3D alignment accuracy.

Graduate Research Student: (ML & Robotics) January 2024 – August 2025

University of Waterloo, RoboHub Lab [Publication, Video, GitHub] Waterloo, Canada

- Developed novel distributed volumetric SLAM in C++/ROS2, integrating TSDF-octree mapping with ORB-SLAM3 localization.
- Fused 4D mmWave radar with RGB-D perception, achieving 92% object-detection accuracy in low-visibility conditions.
- Improved 3D map coverage by 85% over single-robot baselines using coordinated multi-robot frontier-based exploration.

MITACS Industry Thesis Research Partners:

Keysight Inc. [Publication, Video, GitHub] Thesis Internship

- Designed OMPL/RRT*-based collision-free motion-planning for 7-DoF Franka collaborative manipulators.
- Compared PPO reinforcement learning with sampling-based RRT* on trajectory cost, smoothness, and computation time.
- Achieved 100% joint-limit compliance and zero collisions in sim-to-real transfer across all radar-scan trajectories.

Rogers Canada Inc. Thesis Internship

- Deployed 5G-connected multi-robot SLAM and perception for indoor/outdoor AMR mapping, reducing latency 37% over WiFi.
- Evaluated 5G sub-6 GHz and mmWave links for real-time distributed map fusion across cooperative mobile robot teams.

Asset Integrity Engineer: (ML & Robotics) May 2022 – December 2023

ABYSS Solutions Inc. Islamabad, Pakistan (Onsite) & Remote Australia

- Built a Visual-SLAM pipeline for underwater ROV localization and 3D reconstruction of subsea infrastructure.
- Trained U-Net segmentation models for automated corrosion and structural defect detection on ROV-captured imagery.
- Led 10 engineers delivering end-to-end robotic inspection of 18,000+ offshore assets on BP's Clair-Ridge platform.

EDUCATION

University of Waterloo — Waterloo, Canada
Master of Applied Science (MASc), Mechatronics Engineering **(CGPA: 93.5%)** — January 2024 – August 2025

- Thesis link:"Intelligent Multi-Robot Autonomy with Connected AMRs and Manipulators for SMART Factory(s)", Transcript
- **Advisor(s):** Dr. William Melek and Dr. George Shaker. Defence committee: Dr. Soo Jeon, and Dr. Gennaro Notomista

Bahria University — Islamabad, Pakistan
Bachelor of Science (BSc), Electrical (Electronics) Engineering **(CGPA: 80.5%)** — September 2017 – August 2021

- Thesis link, Video: "Swarm Autobots", Transcript
- **Advisor:** Dr. Saleem Aslam Defense committee: Dr. Adil Raja

PUBLICATIONS:

[J2] M. Qureshi, S. Aslam, and G. Shaker, "Swarm Autobots: Pattern Formation Control for Robot Teams," IEEE/ASME Transactions on Mechatronics (TMECH). (Under Review)

[J1] M. Qureshi, M. O. Bagheri, A. Elbadrawy, W. Melek, and G. Shaker, "RAPTAR: Radar Radiation Pattern Acquisition through Automated Collaborative Robotics," IEEE Robotics and Automation Letters (RA-L). (Under Revision) [arXiv]

[C2] M. Qureshi, M. O. Bagheri, W. Melek, and G. Shaker, "Hemispherical Angular Power Mapping of Installed mmWave Radar Modules Under Realistic Deployment Constraints," IEEE APS 2026. (Published) [IEEE Xplore]

[C1] M. Qureshi, A. Werner, Z. Liu, A. Khajepour, G. Shaker, and W. Melek, "CRADMap: Applied Distributed Volumetric Mapping with 5G-Connected Multi-Robots and 4D Radar Perception," IEEE ICARM 2025. (Published) [IEEE Xplore]

SIDE PROJECTS

- **Diffusion Policy vs Imitation learning in Navigation:** [Link] benchmarked diffusion-policy, behavior-cloning, and LSTM imitation learning for TurtleBot 4 navigation; added YOLOv11 for real-time safety detection.
- **Humanoid Robotics: Non-Verbal Human-Robot Interaction** [Link]: Designed a non-verbal gesture-recognition model for multimodal control of a NAO humanoid robot assisting in collaborative breakfast preparation.
- **UAS Challenge 2022, IMechE UK: Aerial Robotics** [Link]: Team Supervisor for a hexacopter UAV (6.9 kg takeoff weight, 100 m operating altitude) for locust-swarm spraying, with a custom carbon-fiber S950 frame and 340 KV BLDC motors.

HONORS & AWARDS

- **Full Graduate Funding Award**, Mitacs Canada & University of Waterloo (MASc) — 2024–2025
- **Top 3 Undergraduate Thesis Project Prize**, Bahria University Video — 2021
- **Star of the Month (Best Performer) Award**, Abyss Solutions Inc. [Link] — March 2023
- **3× Hackathon Winner**, Abyss Solutions Robotics & VR Innovation Challenges — 2022–2023

TALKS AND PRESENTATIONS

Conference Presentation, IEEE APS 2026 — Jul. 2026 Detroit, Michigan, USA

Conference Presentation, IEEE ICARM 2025 — Aug. 2025 Portsmouth, United Kingdom

Symposium Presentation: RoboHub Symposium, Poster & Research Talk — Jan. 2025 University of Waterloo

Conference Presentation, Rogers "Beyond Connectivity" Summit — Jul. 2024 Brampton, Canada

TA, University of Waterloo — May 2024 – August 2025
ME 123 (Electrical Eng for Mechanical Eng), ME 360 (Automatic Control Systems), MTE 201 (Advanced Calculus)

PEER REVIEW

- **Reviewer, IEEE International Conference on Robotics and Automation (ICRA) 2026** (Vienna, Austria; Submission #1761): "ROR-SLAM: UWB Radar-based Object Recognition for Feature-based SLAM."
- **Reviewer, IEEE Sensors Journal** (Manuscript Sensors-69555-2024): "EAMT-SLAM: Edge-Aided Monocular Thermal Sensor SLAM in Low-Illumination Environments."
- **Reviewer, MEIE 2026** (9th Int'l Conference on Mechanical, Electric, and Industrial Engineering; Manuscript MEIE99621): "Feasibility Analysis and Reinforcement Design of a Portable Radar for Use in Vehicle Environments." Recommended minor revisions.